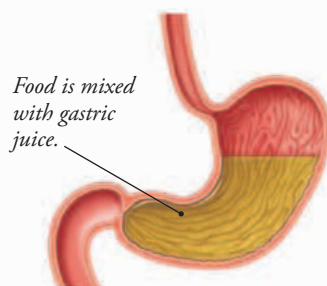


In the stomach

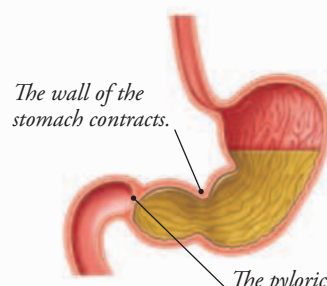
The stomach is a muscular “bag” that sits just below the diaphragm muscle in the middle of the chest. It has two key roles. When you eat, it releases an acidic gastric juice that starts to break down the proteins in food. It also stores food, releasing it at a steady rate so that the small intestine has time to digest it efficiently.

FILLING AND EMPTYING

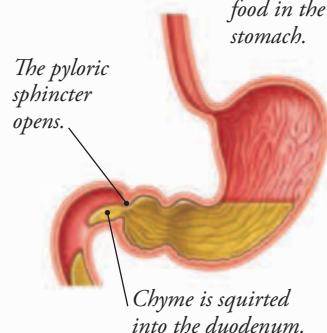
The process of filling and emptying the stomach depends on the type of food eaten, but it takes at least three hours. During this time, food is partially digested and made ready for the main part of digestion in the small intestine. Only when the meal has been turned into liquid form is it released into the duodenum.



◀ **DURING A MEAL**
As the stomach fills and expands, recently chewed food is mixed with gastric juice by contractions of the stomach's wall.



◀ **1-2 HOURS AFTER EATING**
Food is partly digested by gastric juice and churned by powerful muscular contractions into a creamy liquid called chyme.



◀ **3-4 HOURS AFTER EATING**
The pyloric sphincter opens slightly and the stomach wall contracts to push small quantities of chyme into the duodenum, which is the first part of the small intestine.

Within this area is the cardiac sphincter. It stops food from flowing back up into the esophagus.

◀ PYLORIC SPHINCTER

A sphincter is a ring of strong muscle. The pyloric sphincter, seen here through an internal camera, remains tightly closed as the stomach processes food.

Folds in the muscular walls disappear as the stomach expands.

SQUEEZE AND CHURN

The stomach wall has three layers of muscles, arranged at different angles to each other. These produce powerful contractions that squeeze the food and mix it with gastric juice. An enzyme called pepsin gets to work on food proteins, breaking them down into simpler substances.

Chyme (liquid food)